

भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI**

**CS 306: Machine Learning Lab**
**Practice Assignment 3**

**Instructions:** This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. (a) Create a synthetic dataset of 100 patterns to predict sales revenue of a company ($y$) based on the monthly advertising expenditure ($x_1$).

   (b) Split the dataset into training set: testing set with the ratio as below:
   $training$ = 20:20:80 (i.e., initial: 20%, increment by 20%, maximum is 80%)
   $testing$ = 80:20:20 (i.e., initial: 20%, decrement by 20%, minimum is 20%).

   (c) Train simple linear regression models for each training: testing cases from 1(b) using ordinary least square (OLS) method. Save the regression coefficients for each case separately.

   (d) Visualize and store the different plots (best-fit-line) for the different hypotheses designed/ obtained ($\hat{y}$).

   (e) Calculate and store the prediction of salary for the testing patterns over different test splits considering the respective hypothesis.

   (f) Calculate the residual sum of squares (RSS), total sum of squares ($SS_{total}$), and coefficient of determination ($R^2$) for different test splits separately.

   (g) Calculate and analysis the value of Pearson correlation coefficient ($r_{xy}$) from slope of different hypotheses over different training and testing splits, separately.

   (h) What does a high $R^2$-value infer? Does it establish linear correlation between $y$ and $x_1$. Justify your answer from 1(f) and 1(g).

   (i) Report and interpret the results of 1(c)-1(h) in text file. Store the plots in a folder.

2. Execute the above assignment (in question number 1) with inbuilt package for OLS.